

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

EcoTrack AI Smart Plastic Waste Monitoring and Collection Platform

Version: 1.0

Prepared for: Smart India Hackathon (SIH)

Project Type: AI-powered web/mobile waste management platform

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of EcoTrack AI. The system is designed to help citizens report plastic waste, use AI to analyze uploaded images, record geographical locations, manage collection tasks, and provide an administrative dashboard for monitoring and verification.

1.2 Scope

EcoTrack AI provides an end-to-end workflow from plastic-waste reporting to collection verification. The platform can be used by citizens, administrators, and collection teams.

1.3 Objectives

- Enable users to report plastic waste using photographs and location information.
- Use AI/image analysis to identify and classify plastic waste.
- Display reported waste locations on an interactive map.
- Prioritize and assign collection tasks.
- Track reports through Reported, Assigned, In Progress, Collected, and Verified states.
- Provide dashboards and basic analytics for waste-management monitoring.
- Maintain records of reports, users, assignments, and verification evidence.

2. Overall Description

2.1 Product Perspective

EcoTrack AI is a centralized platform consisting of a user interface, backend API, database, AI image-analysis component, mapping/location service, and administrator dashboard.

2.2 User Classes

- Citizen/User: submits waste reports, views report status, and receives updates.

- Administrator: reviews reports, manages users, assigns collection tasks, and monitors analytics.
- Collection Team: views assigned tasks, updates collection status, and uploads completion evidence.

2.3 Assumptions and Constraints

- Users require internet access for online reporting and synchronization.
- Location permission is required for automatic GPS-based reporting.
- AI detection accuracy depends on image quality and training data.
- Map, authentication, storage, and notification services may depend on third-party APIs.
- The prototype may use simulated or limited datasets during SIH development.

3. System Features and Functional Requirements

FR-01 User Registration and Login

The system shall allow users to register and securely log in. Authentication shall distinguish user roles where applicable.

FR-02 Waste Report Creation

The system shall allow a user to upload a waste image and submit a report with location and optional description.

FR-03 AI Plastic Detection

The system shall analyze the uploaded image and return an estimated classification indicating whether plastic waste is present.

FR-04 Waste Classification

The system should classify detected waste into categories such as bottles, bags, wrappers, containers, or other plastic waste when supported by the model.

FR-05 GPS Location

The system shall capture or accept the geographical coordinates of a report and associate them with the report record.

FR-06 Map Visualization

The system shall display report locations on a map and provide basic filtering by status/category.

FR-07 Admin Dashboard

The system shall provide administrators with a dashboard showing reports, statuses, locations, categories, and collection progress.

FR-08 Report Review

Administrators shall be able to review submitted reports and update their status.

FR-09 Task Assignment

Administrators shall be able to assign eligible reports to collection team members.

FR-10 Collection Tracking

Collection teams shall be able to view assigned tasks and update progress.

FR-11 Completion Evidence

The collection team should be able to upload an after-collection image or other verification evidence.

FR-12 Verification

Administrators shall be able to verify or reject completion evidence.

FR-13 Notifications

The system should provide notifications or status messages when important report or task states change.

FR-14 Search and Filtering

The dashboard shall support searching and filtering reports by status, category, date, and location where applicable.

FR-15 Analytics

The system shall provide basic statistics such as total reports, collected reports, pending reports, and category-wise counts.

FR-16 Data Management

The backend shall store report, user, assignment, AI-result, and verification data with appropriate relationships.

4. Non-Functional Requirements

Performance

Normal dashboard and report operations should respond within an acceptable time under the expected prototype load. AI inference time should be minimized.

Security

Passwords and authentication credentials shall be handled securely. Role-based access shall prevent unauthorized administrative actions.

Privacy

The system should collect only necessary user and location data and should protect stored personal information.

Availability

The platform should remain available during normal operating conditions and handle temporary service failures gracefully.

Usability

The user interface shall be simple, mobile-friendly, and understandable for first-time users.

Scalability

The architecture should allow additional users, reports, collection teams, and AI models to be added without major redesign.

Maintainability

The application should use modular frontend, backend, database, and AI components.

Reliability

The system should validate uploads and inputs and avoid losing successfully submitted reports.

Accessibility

Important actions and information should use readable text, clear controls, and responsive layouts.

5. External Interface Requirements

5.1 User Interface

- Responsive reporting interface for mobile and desktop devices.
- Login/registration interface.
- Waste-report form with image upload and location.
- Interactive map showing reported locations.
- Admin dashboard with tables, filters, charts, and status controls.
- Collection-team task interface.

5.2 Software Interfaces

- Backend REST API for communication between frontend and services.
- Database for persistent storage.
- AI/image-classification service or model.
- Map/GPS service for location visualization.
- Optional notification service for status updates.

5.3 Hardware Interfaces

- Smartphone or computer with camera/image-upload capability.
- GPS-capable device for automatic location capture where available.
- Server/cloud infrastructure for application and database hosting.

6. Suggested System Architecture

User/Collection Team/Admin UI → Backend REST API → Authentication & Business Logic
→ Database / File Storage; Backend → AI Detection Service; Backend → Maps/GPS
Service; Backend → Notification Service.

7. Data Requirements

- User: user_id, name, contact/login identifier, role, authentication metadata.
- Waste Report: report_id, user_id, image reference, AI result, category, latitude, longitude, description, timestamp, status.
- Assignment: assignment_id, report_id, collector_id, assigned_at, task status.
- Verification: verification_id, report_id, evidence image reference, verified_by, verification status, timestamp.
- Analytics: derived counts and trends from stored reports and collection records.

8. Report Status Workflow

Reported → AI Analyzed → Reviewed → Assigned → In Progress → Collected → Verified

An administrator may reject invalid, duplicate, or insufficient reports. The exact workflow can be simplified for the SIH prototype.

9. Use Cases

ID	Use Case	Actor	Description
UC-01	Submit Waste Report	User	User uploads an image, provides/permits location, and submits the report.
UC-02	Analyze Image	AI Service	System analyzes the image and stores the detection result.
UC-03	Review Report	Administrator	Admin checks report details and accepts, rejects, or flags the report.
UC-04	Assign Collection	Administrator	Admin assigns an accepted report to a collection team member.
UC-05	Update Collection	Collection Team	Collector views the task and updates its progress.
UC-06	Verify Collection	Administrator	Admin reviews completion evidence and verifies the task.
UC-07	Monitor Analytics	Administrator	Admin views statistics, trends, categories, and collection performance.

10. Recommended Technology Stack

- Frontend: HTML, CSS, JavaScript or React.
- Backend: Python Flask/FastAPI or Node.js/Express.
- Database: Firebase/Firestore, PostgreSQL, or MongoDB.
- AI: Python-based computer-vision model such as a suitable object-detection/classification model.
- Maps: Google Maps, OpenStreetMap, or another suitable mapping API.
- Hosting: Vercel for frontend and a suitable cloud service for backend/API and AI inference.

11. Security Requirements

- Use secure authentication and authorization.
- Validate uploaded files and input data.
- Restrict admin and collection-team functions by role.
- Protect API endpoints against unauthorized access.
- Use HTTPS in production.
- Avoid exposing private API keys in frontend source code.

12. Future Enhancements

- Real-time nearby collector assignment and route optimization.
- Advanced AI detection for multiple waste categories and severity estimation.
- Duplicate-report detection.
- Heatmaps for high-waste areas.
- Reward or gamification system for verified citizen reports.
- Municipal/NGO integration and automated reporting.
- Predictive analytics for waste hotspots.

13. Acceptance Criteria

- A registered user can submit a waste image and location successfully.
- The system produces and stores an AI analysis result.
- The submitted report appears on the admin dashboard and map.
- An administrator can review and assign a report.
- A collection team member can update task status.
- Completion evidence can be submitted and verified.
- Dashboard statistics reflect stored report and collection data.
- Unauthorized users cannot access protected administrative functions.

14. Conclusion

EcoTrack AI aims to transform plastic-waste reporting from a simple complaint mechanism into a measurable, AI-assisted workflow. By connecting image analysis, geolocation,

administration, collection assignment, and verification, the platform provides a practical foundation for a scalable smart waste-management solution.